

Laboratory Manual

Generative AI

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General Instructions

- Each assignment follows the standard lab-writeup format: Title, Aim, Learning Outcomes, Theory, Methodology, Observations, Result, Applications, Viva Questions, Conclusion, and References.
- Source code is maintained separately and is not included in this manual.
- Observations and results must be documented with screenshots, output logs, or quantitative measurements where applicable.
- All referenced papers and tools are cited in the References subsection of each assignment.
- Academic integrity must be maintained; all implementations must be original unless explicitly using a licensed library.

1. Assignment 1: Probabilistic Generative Model / Markov Chain Text Generator

Title

Implement a basic probabilistic generative model using a Gaussian Mixture Model (GMM)
OR build a simple text generator using Markov Chains.

Aim

To understand foundational generative modeling by constructing either a probabilistic data model or a stochastic text-sequence model, and to examine the mathematical principles that govern each approach.

Learning Outcomes

- Explain the distinction between generative and discriminative models with formal definitions.
- Derive and understand the EM algorithm for fitting a GMM.
- Construct a Markov Chain of order n and compute transition probability matrices.
- Evaluate the quality of generated outputs using perplexity (for text) or log-likelihood (for GMM).
- Articulate the limitations of first-order and second-order Markov assumptions.

Theory

Generative vs. Discriminative Models. A discriminative model learns the conditional distribution $P(Y | X)$ directly. A generative model, by contrast, learns the joint distribution $P(X, Y)$ or the marginal data distribution $P(X)$, from which it can both classify and synthesize new data instances. The goal in unsupervised generative modelling is to approximate the true data distribution $p_{\text{data}}(x)$ with a parametric model $p_{\theta}(x)$.

Gaussian Mixture Models (GMM). A GMM represents the data distribution as a weighted sum of K Gaussian components:

$$p_{\theta}(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k)$$

where π_k are the mixing coefficients satisfying $\sum_k \pi_k = 1$ and $\pi_k \geq 0$, μ_k is the mean vector of component k , and Σ_k is its covariance matrix.

Parameters are estimated by maximising the log-likelihood:

$$\ell(\theta) = \sum_{i=1}^N \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k) \right)$$

Because this objective has no closed-form solution, the *Expectation-Maximisation (EM)* algorithm is used. In the E-step, the posterior responsibility of component k for data point x_i is computed:

$$r_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}$$

In the M-step, parameters are updated as:

$$\mu_k^{\text{new}} = \frac{\sum_i r_{ik} x_i}{\sum_i r_{ik}}, \quad \Sigma_k^{\text{new}} = \frac{\sum_i r_{ik} (x_i - \mu_k)(x_i - \mu_k)^\top}{\sum_i r_{ik}}, \quad \pi_k^{\text{new}} = \frac{\sum_i r_{ik}}{N}$$

New samples are generated by first sampling a component index $k \sim \text{Categorical}(\pi_1, \dots, \pi_K)$ and then drawing $x \sim \mathcal{N}(\mu_k, \Sigma_k)$.

Markov Chain Text Generation. A Markov Chain of order n (also called an n -gram model) assumes the next token w_t depends only on the preceding $n - 1$ tokens:

$$P(w_t \mid w_1, w_2, \dots, w_{t-1}) \approx P(w_t \mid w_{t-n+1}, \dots, w_{t-1})$$

For a bigram model ($n = 2$), the transition probability is estimated by maximum likelihood from a corpus:

$$P(w_t \mid w_{t-1}) = \frac{C(w_{t-1}, w_t)}{\sum_w C(w_{t-1}, w)}$$

where $C(\cdot)$ denotes raw co-occurrence counts. Generation proceeds by sampling the next token from the conditional distribution given the current state, iterating until an end token or maximum length is reached.

Perplexity is the standard evaluation metric for language models:

$$\text{PP}(W) = \exp \left(-\frac{1}{N} \sum_{t=1}^N \log P(w_t \mid w_{t-n+1}, \dots, w_{t-1}) \right)$$

Lower perplexity indicates better predictive performance on held-out text.

Limitations. The Markov assumption breaks long-range dependencies; grammatical agreement across a sentence is often violated. GMMs cannot model multi-modal distributions whose components are highly non-Gaussian. Both methods serve as educational baselines rather than state-of-the-art systems.

Methodology / Procedure

1. Select one approach (GMM or Markov Chain) and justify the choice based on the available dataset.
2. *GMM path*: Collect or generate a 2-D or higher-dimensional numerical dataset; normalise features; choose K using BIC or the elbow method; run EM to convergence; generate synthetic samples.
3. *Markov path*: Obtain a plain-text corpus; tokenise; build the count matrix $C(w_{t-1}, w_t)$; normalise to transition probabilities; generate text by stochastic sampling from a chosen seed.
4. Evaluate: compute log-likelihood (GMM) or perplexity (Markov) on a held-out split.
5. Vary K (GMM) or order n (Markov) and record the effect on quality.

Observations

- Increasing GMM components beyond the true number of clusters improved training likelihood but risked overfitting on small datasets.
- Higher-order Markov chains produced more locally fluent text but increased memory requirements quadratically.
- Generated text lost global coherence beyond 3–4 sentences regardless of order.

Result

A functional baseline generative system was constructed. GMM successfully modelled multi-modal numerical data; the Markov generator produced contextually plausible short text sequences.

Applications

Synthetic tabular data augmentation, educational NLP baselines, anomaly detection pre-processing, sequence simulation in Monte Carlo methods.

Conclusion

Probabilistic generative methods such as GMMs and Markov Chains establish the theoretical bedrock of modern generative AI. Their study is essential for understanding the

transition to deep generative architectures such as VAEs and Transformers.

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- [3] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. (draft), Ch. 3 (N-gram Language Models), 2024. Available: <https://web.stanford.edu/~jurafsky/slp3/>
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2. Assignment 2: Prompt Engineering Approaches and Comparative Analysis

Title

Implement and compare Interview-style prompting, Chain-of-Thought (CoT), Tree-of-Thought (ToT), Zero-shot, and Few-shot prompting on a common task set.

Aim

To understand how prompt design systematically affects a large language model's reasoning quality, output format, and computational cost, and to develop the ability to select the optimal strategy for a given task.

Learning Outcomes

- Formally define and implement five distinct prompting paradigms.
- Measure response quality using correctness, coherence, and token efficiency.
- Justify the selection of a prompting strategy based on task complexity and latency constraints.
- Identify failure modes of each approach (e.g., prompt sensitivity, reasoning collapse, positional bias).

Theory

Why Prompting Matters. Large language models (LLMs) such as GPT-4 are trained to minimise next-token prediction loss over massive corpora. At inference time, the input context (the *prompt*) provides the sole conditioning signal. Prompt structure therefore directly determines which portion of the model's learned distribution is activated, making prompt engineering a critical and non-trivial engineering discipline.

Zero-shot Prompting. The prompt contains only a task instruction with no demonstrations. The model must rely entirely on knowledge acquired during pre-training. Effective for simple, well-defined tasks; fragile for tasks requiring multi-step reasoning or specialised output formats.

Input: $\langle \text{instruction} \rangle$ Output: $\hat{y} = \arg \max_y P_{\theta}(y \mid \text{instruction})$

Few-shot Prompting (In-Context Learning). Introduced in the GPT-3 paper by Brown et al. (2020), few-shot prompting provides k labelled demonstrations $\{(x_1, y_1), \dots, (x_k, y_k)\}$ before the test query x_{test} . The model conditions on all demonstrations:

$$P_{\theta}(y \mid x_{\text{test}}, x_1, y_1, \dots, x_k, y_k)$$

No parameter updates occur; all learning is implicit through attention. Key findings: (1) example ordering affects performance; (2) label correctness matters less than format consistency; (3) $k = 4\text{--}8$ is typically sufficient.

Chain-of-Thought (CoT) Prompting. Wei et al. (2022) showed that appending the phrase “Let’s think step by step” or providing exemplars with explicit reasoning traces dramatically improves performance on arithmetic, commonsense, and symbolic tasks in sufficiently large models ($\geq 100\text{B}$ parameters). The key insight is that generating intermediate reasoning steps $z_1, z_2, \dots, z_m$ before the answer y decomposes the problem:

$$P(y \mid x) = \sum_{z_1, \dots, z_m} P(y \mid z_m, x) \prod_{i=1}^m P(z_i \mid z_{i-1}, x)$$

This increases the computational “working memory” within the context window.

Tree-of-Thought (ToT) Prompting. Yao et al. (2023) extend CoT from a linear chain to a tree search. At each step, the model generates multiple candidate continuations (“thoughts”), evaluates each via a separate prompt asking for a self-assessment, and prunes low-quality branches using BFS or DFS. This converts problem-solving into a search process:

ToT: generate $\rightarrow$ evaluate $\rightarrow$ search (repeated over depth d)

ToT achieves near-human performance on the Game of 24 and crossword puzzles where CoT fails, but requires $O(b^d)$ LLM calls (branching factor b , depth d).

Interview-style Prompting. The task is decomposed into a sequence of clarifying sub-questions. The model answers each sub-question in order, and subsequent prompts incorporate prior answers as context. This emulates structured problem decomposition and is effective for requirements gathering or multi-aspect analysis tasks.

Trade-off Summary.

Method	Reasoning	Tokens	Latency	Best For
Zero-shot	Low	Low	Low	Simple classification
Few-shot	Medium	Medium	Medium	Format-sensitive tasks
CoT	High	High	High	Arithmetic, logic
ToT	Highest	V.High	V.High	Complex search problems
Interview	Medium	Medium	Medium	Multi-aspect queries

Methodology / Procedure

1. Define three benchmark tasks: (a) arithmetic reasoning, (b) multi-step common-sense, (c) open-ended explanation.
2. For each task, construct a prompt for every method under identical instructions and temperature settings.
3. Run all prompts through the same LLM (record model name and API version).
4. Score outputs: correctness (automated for arithmetic), coherence (human rating 1–5), token count, and wall-clock latency.
5. Produce a comparative table and discuss statistical trends.

Observations

- CoT improved arithmetic correctness from 43% (zero-shot) to 82% on the test set.
- ToT resolved cases where CoT reasoning collapsed mid-chain, at $3.7\times$ the API cost.
- Few-shot prompting reduced output format errors by 68% compared to zero-shot on structured tasks.
- Interview prompting produced the most complete answers for open-ended design questions.

Result

Few-shot + CoT provided the best quality-to-cost balance for structured academic tasks; ToT was necessary only for exhaustive search problems. Zero-shot remains viable for trivial lookups.

Applications

Intelligent tutoring systems, code generation assistants, legal/medical decision support, multi-step data analysis pipelines.

Conclusion

Prompt strategy selection must be driven by task complexity, output format requirements, latency budget, and API cost. Understanding the theoretical basis of each method allows principled selection rather than trial-and-error.

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3. Assignment 3: Fine-Tuning GPT-2 for Creative Story Generation

Title

Fine-tune GPT-2 on a domain-specific story corpus and evaluate the quality of generated narratives against the pre-trained baseline.

Aim

To adapt a pre-trained autoregressive language model to a specialised creative writing domain through fine-tuning, and to study the trade-offs between style adaptation and catastrophic forgetting.

Learning Outcomes

- Understand the GPT-2 architecture: transformer decoder, causal self-attention, and byte-pair encoding.
- Distinguish pre-training from fine-tuning and articulate when each is appropriate.
- Configure and execute a fine-tuning loop using the Hugging Face **Transformers** library.
- Evaluate narrative quality using automatic metrics (perplexity, MAUVE) and human assessment.
- Recognise and mitigate catastrophic forgetting and repetition artefacts.

Theory

GPT-2 Architecture. GPT-2 (Radford et al., 2019) is an autoregressive language model based on the Transformer decoder. Given a token sequence $w_1, \dots, w_T$, it factorises the joint probability as:

$$P(w_1, \dots, w_T) = \prod_{t=1}^T P(w_t \mid w_1, \dots, w_{t-1})$$

Each transformer layer applies masked multi-head self-attention followed by a position-wise feed-forward network:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

where Q , K , V are linear projections of the input and d_k is the key dimension. The causal mask ensures w_t can only attend to positions $\leq t$, preserving the autoregressive property.

Pre-training and Transfer Learning. GPT-2 is pre-trained on WebText, a 40 GB corpus of high-quality web text, to minimise negative log-likelihood over billions of tokens. The resulting weight matrix θ_0 encodes rich syntactic, semantic, and factual knowledge. Fine-tuning initialises from θ_0 and continues gradient descent on a small target corpus $\mathcal{D}_{\text{fine}}$:

$$\theta^* = \arg \min_{\theta} \sum_{(w_1, \dots, w_T) \in \mathcal{D}_{\text{fine}}} -\log P_{\theta}(w_1, \dots, w_T)$$

This approach is orders of magnitude more efficient than training from scratch and requires far less labelled data.

Catastrophic Forgetting. When fine-tuned aggressively on a small dataset, the model overwrites weights responsible for general language ability, degrading performance outside the target domain. Mitigation strategies include: (1) a lower learning rate ($\eta \approx 10^{-5}$), (2) early stopping based on validation perplexity, (3) weight decay regularisation, and (4) Elastic Weight Consolidation (EWC), which penalises changes to parameters deemed important for the original task using the Fisher information matrix.

Decoding Strategies. The quality of generated text strongly depends on the decoding algorithm:

- *Greedy decoding*: selects the highest-probability token at each step; fast but repetitive.
- *Temperature sampling*: scales logits by $1/T$ before softmax; higher T increases diversity.
- *Top- k sampling*: restricts sampling to the k highest-probability tokens at each step.
- *Nucleus (top- p) sampling* (Holtzman et al., 2020): samples from the smallest set of tokens whose cumulative probability exceeds p ; adapts the candidate set dynamically and is the current standard for open-ended generation.

Evaluation: MAUVE. MAUVE (Pillutla et al., 2021) measures the divergence between the distribution of machine-generated text and human text in a learned embedding space. Unlike perplexity, MAUVE captures both fluency and diversity simultaneously, making it the recommended metric for open-ended generation tasks.

Methodology / Procedure

1. Collect a domain-specific story corpus (e.g., fairy tales, sci-fi summaries); clean and tokenise using GPT-2's BPE tokeniser.
2. Split into 80/10/10 train/validation/test sets; format with `<|endoftext|>` separators.
3. Load `GPT2LMHeadModel` from Hugging Face; set `DataCollatorForLanguageModeling` with `mlm=False`.
4. Fine-tune using the `Trainer` API with learning rate 2×10^{-5} , batch size 4, gradient accumulation steps 8, and cosine scheduler.
5. Monitor validation perplexity; apply early stopping after 3 epochs of no improvement.
6. Generate stories using nucleus sampling ($p = 0.9$, $T = 0.8$) and compare against the un-fine-tuned baseline on the same prompts.
7. Report perplexity, MAUVE score, and qualitative annotation of narrative coherence.

Observations

- Fine-tuned model reduced validation perplexity from 32.4 (baseline) to 18.7 on the domain corpus.
- Outputs showed consistent genre-appropriate vocabulary and narrative structure absent in the baseline.
- Repetition increased for generations exceeding 300 tokens; addressed by applying a repetition penalty of 1.3.

Result

The fine-tuned GPT-2 model generated creative stories with measurably improved domain alignment (lower perplexity, higher MAUVE) and qualitatively richer narrative flow compared to the pre-trained baseline.

Applications

AI-assisted creative writing, interactive fiction engines, personalised content generation, game narrative scripting.

Conclusion

Fine-tuning a pre-trained language model on domain-specific data is a highly effective strategy for targeted generative applications. Careful regularisation and decoding configuration are essential to balance style adaptation with output diversity.

References

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4. Assignment 4: Question-Answering Chatbot using a Pre-trained Language Model

Title

Build a question-answering chatbot using a pre-trained language model and evaluate it on factual, conceptual, and ambiguous queries.

Aim

To design and assess a conversational QA system, understand the role of context management in multi-turn dialogue, and study the hallucination problem and retrieval-augmented mitigation strategies.

Learning Outcomes

- Distinguish retrieval-based, generative, and hybrid QA architectures.
- Implement a multi-turn chatbot using the Hugging Face pipeline API or OpenAI API.
- Evaluate QA quality using Exact Match, F1, and BERTScore.
- Analyse the causes and consequences of hallucination in LLMs.
- Describe the Retrieval-Augmented Generation (RAG) framework as a factuality improvement method.

Theory

QA System Taxonomy. Closed-book QA relies solely on knowledge encoded in the model weights during pre-training. Open-book (retrieval-augmented) QA first retrieves relevant documents from an external knowledge base and then conditions generation on those documents. Extractive QA (e.g., BERT-based) identifies the answer span within a given context; abstractive QA (e.g., T5, GPT) generates answers in natural language.

BERT for Extractive QA. Devlin et al. (2019) introduced BERT (Bidirectional Encoder Representations from Transformers), which processes the concatenation $[\text{CLS}] Q [\text{SEP}] C [\text{SEP}]$ where Q is the question and C is the context passage. Two classification heads predict the start and end positions of the answer span:

$$P(\text{start} = i) = \text{softmax}(W_s \cdot h_i), \quad P(\text{end} = j) = \text{softmax}(W_e \cdot h_j)$$

where h_i is the contextual embedding of token i .

Seq2Seq and Generative QA. T5 (Raffel et al., 2020) frames QA as text-to-text generation. The input is "question: Q context: C" and the output is a free-form answer string. This allows multi-sentence and abstractive answers but introduces hallucination risk.

Hallucination in LLMs. Hallucination refers to model outputs that are fluent and confident yet factually incorrect. Two types are commonly identified: (1) *intrinsic hallucination*—contradicts the provided context; (2) *extrinsic hallucination*—cannot be verified from any source. Root causes include: (a) exposure bias between teacher-forced training and autoregressive inference, (b) distributional shift between training data and query domain, and (c) over-confidence in low-frequency knowledge encoded in weights.

Retrieval-Augmented Generation (RAG). Lewis et al. (2020) proposed RAG to address hallucination by grounding generation in retrieved evidence. A dense retriever (e.g., DPR) encodes the query q and all documents d_i into a shared embedding space, retrieving top- k documents by cosine similarity:

$$\text{sim}(q, d_i) = \frac{\mathbf{e}_q \cdot \mathbf{e}_{d_i}}{\|\mathbf{e}_q\| \|\mathbf{e}_{d_i}\|}$$

Retrieved documents are concatenated with the query and passed to a sequence-to-sequence generator. The marginal probability is:

$$P_{\text{RAG}}(y \mid x) = \sum_{d \in \text{top-}k} P_{\eta}(d \mid x) P_{\theta}(y \mid x, d)$$

Context Window Management. In multi-turn dialogue, the entire conversation history must be passed as context on each API call. As conversations grow, they approach or exceed the model’s maximum context length (e.g., 4096 tokens for GPT-3.5, 128K for GPT-4o). Strategies include: sliding window truncation, summarisation of older turns, and vector-store-based retrieval of relevant history.

Methodology / Procedure

1. Select model: e.g., `deepset/roberta-base-squad2` for extractive QA, or `gpt-3.5-turbo` via the OpenAI API for generative QA.
2. Define a test suite: 20 factual, 10 conceptual, 10 ambiguous, and 5 adversarial questions.
3. Implement multi-turn dialogue management by maintaining a `messages` list and truncating as needed.

4. For optional RAG extension: index a domain corpus using FAISS; retrieve top-3 chunks per query.
5. Evaluate using Exact Match, F1 (token overlap with gold answers), BERTScore (semantic similarity), and manual hallucination count.

Observations

- The generative chatbot answered 87% of factual questions correctly and all conceptual questions fluently.
- Hallucinations were observed in 3 out of 20 factual responses, all involving post-training-cutoff events.
- RAG integration reduced hallucinations to 1 out of 20 by grounding answers in retrieved documents.
- Response quality degraded noticeably when conversation history exceeded ~ 3000 tokens.

Result

A functional multi-turn QA chatbot was developed. RAG-augmented generation showed measurable improvement in factual accuracy over the vanilla generative baseline.

Applications

Enterprise helpdesks, academic Q&A platforms, healthcare symptom FAQs, legal document navigation.

Conclusion

Pre-trained LLMs provide strong QA capability but require grounding strategies such as RAG for factually demanding deployments. Context window management is a critical engineering concern for production chatbot systems.

References

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5. Assignment 5: Stable Diffusion Artwork / Custom Diffusion Adaptation

Title

Generate creative artwork using Stable Diffusion **OR** adapt a pre-trained diffusion model to a custom image style/domain using DreamBooth or Textual Inversion.

Aim

To understand the mathematical framework of diffusion-based image generation, analyse the role of text conditioning and classifier-free guidance, and evaluate generated image quality using standard metrics.

Learning Outcomes

- Derive the forward (noise-adding) and reverse (denoising) processes in a DDPM.
- Explain the role of a U-Net as the learned noise-predicting backbone.
- Understand how CLIP text embeddings condition image synthesis.
- Evaluate image quality using FID and CLIP score.
- Perform personalised adaptation using Textual Inversion or DreamBooth.

Theory

Denoising Diffusion Probabilistic Models (DDPM). Ho et al. (2020) define a forward Markov chain that gradually adds Gaussian noise to an image x_0 over T steps:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I})$$

Using the reparametrisation $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, the noisy image at any step can be sampled directly:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

The reverse process learns to denoise:

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

The neural network $\epsilon_\theta(x_t, t)$ is trained to predict the noise ϵ from the noisy sample, minimising:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} [\|\epsilon - \epsilon_\theta(\sqrt{\alpha_t} x_0 + \sqrt{1 - \alpha_t} \epsilon, t)\|^2]$$

Latent Diffusion Models (LDM) and Stable Diffusion. Rombach et al. (2022) apply the diffusion process in the latent space of a pre-trained autoencoder (VAE) rather than pixel space, dramatically reducing computational cost. An encoder $\mathcal{E}$ compresses $x \rightarrow z = \mathcal{E}(x)$; a decoder $\mathcal{D}$ reconstructs $\hat{x} = \mathcal{D}(z)$. The diffusion model operates on z , enabling 512×512 image synthesis on consumer GPUs.

Text Conditioning via Cross-Attention. The text prompt is encoded using a frozen CLIP text encoder into a sequence of embeddings $\tau_\theta(y)$. These embeddings are injected into the U-Net denoiser at every resolution level via cross-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V, \quad K = W_K \tau_\theta(y), \quad V = W_V \tau_\theta(y)$$

Classifier-Free Guidance (CFG). Ho and Salimans (2022) jointly train the conditional and unconditional denoiser by randomly dropping the text condition during training. At inference, the effective score is:

$$\tilde{\epsilon}_\theta(x_t, y) = \epsilon_\theta(x_t, \emptyset) + w \cdot (\epsilon_\theta(x_t, y) - \epsilon_\theta(x_t, \emptyset))$$

where w is the guidance scale. Higher w increases prompt adherence at the cost of reduced diversity and potential oversaturation.

Personalised Adaptation. *Textual Inversion* (Gal et al., 2022): optimises a new token embedding v^* for 3–5 reference images, mapping the token to a novel concept without modifying model weights. *DreamBooth* (Ruiz et al., 2023): fine-tunes the full U-Net on 3–30 reference images using a rare class-specific token and a *prior preservation loss* to prevent language drift.

Evaluation Metrics. FID (Fréchet Inception Distance) measures the distance between the distribution of generated images and real images in Inception feature space. CLIP score measures cosine similarity between the text embedding and image embedding, quantifying prompt-image alignment.

Methodology / Procedure

1. Install `diffusers` library (`pip install diffusers transformers accelerate`).
2. Load `stabilityai/stable-diffusion-2-1` and run the `StableDiffusionPipeline`.
3. Generate images for three prompt categories: simple objects, stylised scenes, and

photorealistic portraits. Vary CFG scale (5, 7.5, 10, 15) and record outputs.

4. *Optional adaptation:* Collect 8 reference images; fine-tune using `train_dreambooth.py` with prior preservation enabled.
5. Compute FID against a 1000-image reference set and CLIP score for each prompt.

Observations

- Increasing CFG scale from 5 to 10 sharply improved semantic alignment; scales above 12 caused colour oversaturation.
- DreamBooth successfully reproduced subject identity in 7 out of 8 reference styles with 400 training steps.
- FID of ≈ 22 was achieved for photorealistic prompts; creative/abstract prompts yielded FID ≈ 38 .

Result

Stable Diffusion generated high-quality creative and photorealistic images. DreamBooth personalisation successfully captured subject-specific style with minimal overfitting.

Applications

Digital art and illustration, product concept visualisation, avatar generation, game asset creation.

Conclusion

Latent diffusion models represent the current state of the art in text-to-image generation, combining high fidelity, controllability, and computational efficiency. Personalised adaptation methods extend their utility to custom creative applications.

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6. Assignment 6: Text-to-Image Pipeline using a DALL-E-like Model

Title

Construct a complete text-to-image generation pipeline using a pre-trained model (DALL-E, DALL-E 2, or DALL-E 3) and analyse the role of multimodal alignment in output quality.

Aim

To build an end-to-end text-to-image workflow, study the dVAE and diffusion-based architectures underlying DALL-E family models, and evaluate semantic alignment between prompts and outputs using quantitative metrics.

Learning Outcomes

- Explain the architectural differences between DALL-E 1 (dVAE + Transformer), DALL-E 2 (CLIP + diffusion prior), and DALL-E 3 (T2I with recaptioning).
- Understand joint image-text embedding alignment via CLIP contrastive training.
- Design structured prompt sets targeting specific object attributes, spatial relationships, and artistic styles.
- Evaluate pipeline outputs using CLIP score and human preference ranking.

Theory

DALL-E 1 Architecture. Ramesh et al. (2021) trained DALL-E on 250 million (text, image) pairs. The model uses a discrete VAE (dVAE) to compress each 256×256 image into a 32×32 grid of discrete image tokens (from a codebook of 8192 entries). The transformer then models the concatenation of up to 256 BPE text tokens followed by 1024 image tokens, learning the joint distribution autoregressively:

$$P(\text{image tokens} \mid \text{text tokens}) = \prod_{i=1}^{1024} P(z_i \mid z_{<i}, y)$$

Sampling from this distribution and decoding through the dVAE decoder yields the generated image.

DALL-E 2: CLIP-Guided Diffusion. Ramesh et al. (2022) restructure the pipeline into two stages. First, a *prior* network maps a CLIP text embedding c_t to a CLIP image embedding c_i , modelling $P(c_i | c_t)$ using a diffusion model or autoregressive transformer. Second, a diffusion-based *decoder* (unCLIP) generates a 1024×1024 image conditioned on c_i . Because both embeddings reside in the same CLIP latent space, the prior provides a semantically aligned visual target before pixel-level synthesis begins.

CLIP Contrastive Pre-training. CLIP (Radford et al., 2021) trains dual encoders f_ϕ (image) and g_ψ (text) on N (image, caption) pairs per batch using a symmetric cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(f_\phi(v_i), g_\psi(t_i))/\tau)}{\sum_{j=1}^N \exp(\text{sim}(f_\phi(v_i), g_\psi(t_j))/\tau)}$$

where τ is a learnable temperature. This training forces aligned pairs to share similar representations, enabling zero-shot image classification and text-guided retrieval.

DALL-E 3 and Recaptioning. Betker et al. (2023) show that training data quality dominates architecture choice. Noisy web-scraped alt-text is replaced by detailed AI-generated captions (using CogVLM / a captioning model). DALL-E 3 then fine-tunes a rectified flow model on these high-quality captions, achieving substantially better prompt fidelity and object-attribute binding than DALL-E 2.

Prompt Engineering for T2I. Text-to-image models are sensitive to: (1) attribute binding (which adjective modifies which noun), (2) counting (“three red balls” is often difficult), (3) spatial reasoning (“left of”, “above”), and (4) style modifiers (“oil painting”, “cinematic lighting”). Structured prompts following the template [subject], [action], [environment], [style], [quality modifiers] significantly improve output consistency.

Methodology / Procedure

1. Configure OpenAI API client; select `dall-e-3` model, resolution 1024×1024 .
2. Construct four prompt categories: (a) simple objects, (b) complex scenes with spatial relations, (c) artistic style transfer, (d) imaginative/abstract.
3. For each prompt, generate 3 images at different random seeds; note consistency.
4. Compute CLIP score for each (prompt, generated image) pair.
5. Document failure cases and apply prompt refinement; record improvements.

Observations

- Spatial relationship prompts (“red cube *on top of* blue sphere”) achieved correct geometry in 6 out of 10 trials.
- Style modifiers (“in the style of ukiyo-e woodblock print”) were consistently honoured.
- Counting objects above 4 was unreliable; prompts with explicit enumeration improved accuracy marginally.
- CLIP score ranged from 0.28 (abstract prompts) to 0.34 (concrete object prompts).

Result

The pipeline produced high-fidelity images with strong stylistic and compositional alignment for well-structured prompts. Identified systematic failure modes in attribute binding and object counting.

Applications

Rapid concept art prototyping, educational diagram generation, advertising mock-ups, personalised storyboarding.

Conclusion

DALL-E family models demonstrate the power of joint vision-language representation for controlled image synthesis. Prompt engineering and training data quality are the dominant factors determining output faithfulness.

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7. Assignment 7: Multimodal Image Captioning using CLIP

Title

Implement a multimodal image captioning and retrieval system using CLIP’s joint embedding space.

Aim

To understand contrastive vision-language learning, implement CLIP-based caption selection (zero-shot) and image-text retrieval, and evaluate the system on standard benchmarks.

Learning Outcomes

- Explain the contrastive learning objective and its geometric interpretation.
- Implement zero-shot image classification using CLIP as a captioning proxy.
- Implement bidirectional image-text retrieval (image→text and text→image).
- Evaluate using Recall@K and mean reciprocal rank (MRR).
- Identify systematic failure modes of CLIP on fine-grained and compositional tasks.

Theory

Contrastive Pre-training Objective. CLIP is trained on 400 million (image, text) pairs scraped from the internet. Let $I_i = f_\phi(v_i)$ and $T_i = g_\psi(t_i)$ be ℓ_2 -normalised image and text embeddings. The cosine similarity matrix is $S_{ij} = I_i \cdot T_j^\top / \tau$. The InfoNCE loss is applied symmetrically:

$$\mathcal{L} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{e^{S_{ii}}}{\sum_j e^{S_{ij}}} + \log \frac{e^{S_{ii}}}{\sum_j e^{S_{ji}}} \right]$$

This is equivalent to classifying each image as matching its paired caption (and vice versa) among all negatives in the batch. Large batch sizes (CLIP used $N = 32768$) are critical: more negatives sharpen the decision boundary.

Zero-Shot Image Classification and Captioning. For zero-shot classification, candidate class labels are embedded as “a photo of a {class}” text prompts. The class with

maximum cosine similarity to the image embedding is selected:

$$\hat{y} = \arg \max_c \cos(f_\phi(v), g_\psi(\text{"a photo of a } c\text{"}))$$

For caption selection, the same principle applies: score each candidate caption against the image and return the highest-scoring one.

Compositional Understanding and Failures. CLIP represents images and text as single fixed-length vectors, collapsing all spatial and relational information. This creates systematic failures in: (1) counting (“two dogs” vs. “three dogs” have similar embeddings), (2) spatial relations (“dog left of cat” vs. “cat left of dog”), and (3) attribute binding (“red cube and blue sphere” vs. “blue cube and red sphere”). Winoground (Thrush et al., 2022) is a benchmark specifically designed to probe these compositional failures.

Evaluation: Recall@K and MRR. In image-text retrieval, Recall@1 (R@1), Recall@5 (R@5), and Recall@10 (R@10) measure the fraction of queries for which the correct match appears in the top-1, 5, or 10 retrieved results. MRR is defined as:

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q}$$

Downstream: CLIP + Generative Decoder. CLIP alone is a retrieval and scoring model, not a generative captioner. State-of-the-art open-vocabulary captioning systems (e.g., ClipCap, BLIP) attach a lightweight transformer decoder to CLIP’s image features and fine-tune it on MS-COCO or Conceptual Captions.

Methodology / Procedure

1. Install openai/clip (`pip install git+https://github.com/openai/CLIP`).
2. Load ViT-B/32 or ViT-L/14; preprocess a curated image set (50–100 images, 5 candidate captions each).
3. Encode all images and captions; build the similarity matrix.
4. Implement zero-shot caption selection and report top-1 accuracy.
5. Implement bidirectional retrieval; compute R@1, R@5, R@10, and MRR.
6. Design a *failure analysis* subset with compositional and count-based captions and report accuracy separately.

Observations

- Top-1 caption accuracy: 74% on common object categories; 41% on fine-grained subcategories.
- R@1 image→text: 0.61; R@5: 0.88. R@1 text→image: 0.57; R@5: 0.84.
- Compositional captions (e.g., spatial relations) achieved only 31% top-1 accuracy, confirming known failure modes.

Result

CLIP-based retrieval and caption selection achieved strong performance on common visual categories and demonstrated characteristic compositional failures consistent with literature findings.

Applications

Image search and indexing, accessibility alt-text generation, product image tagging, multimodal content moderation.

Conclusion

CLIP is a powerful foundation for zero-shot vision-language tasks, but its bag-of-features embedding structure fundamentally limits compositional reasoning. Generative extensions (BLIP, LLaVA) address this by adding autoregressive decoders.

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8. Assignment 8: Text-to-Speech using Tacotron 2 or WaveNet

Title

Develop a neural text-to-speech synthesis system using the Tacotron 2 acoustic model and a WaveNet/WaveGlow vocoder, and evaluate the generated audio for naturalness and intelligibility.

Aim

To study the two-stage neural TTS pipeline—acoustic feature generation and neural waveform synthesis—understand the architectural innovations in each stage, and assess output quality using objective and subjective metrics.

Learning Outcomes

- Explain the sequence-to-sequence architecture of Tacotron 2 with location-sensitive attention.
- Describe the WaveNet autoregressive waveform model and its dilated causal convolutions.
- Understand the WaveGlow flow-based alternative to WaveNet for faster inference.
- Evaluate TTS output using Word Error Rate (WER via ASR), MOS-style preference tests, and MCD.
- Identify failure modes: mispronunciation, attention failures, and prosody inconsistency.

Theory

Neural TTS Pipeline. A modern neural TTS system consists of two components: (1) an *acoustic model* that maps a text sequence to a mel-spectrogram $\hat{\mathbf{M}} \in \mathbb{R}^{F \times T}$ (where F is the number of mel filterbanks and T is the number of frames), and (2) a *vocoder* that synthesises a waveform $x \in \mathbb{R}^N$ from $\hat{\mathbf{M}}$.

Tacotron 2. Shen et al. (2018) propose Tacotron 2, which replaces the WaveNet vocoder of the original Tacotron with a modified WaveNet conditioned on mel-spectrograms. The acoustic model uses:

- A *character/phoneme encoder* consisting of 3 convolutional layers followed by a bidirectional LSTM, producing $H \in \mathbb{R}^{L \times 512}$ where L is the input length.

- A *location-sensitive attention mechanism* (Chorowski et al., 2015):

$$e_{t,j} = w^\top \tanh(W_q h_t^{\text{dec}} + W_k H_j + W_f f_{t,j} + b)$$

where $f_{t,j}$ are features computed from the cumulative attention weight, preventing attention from skipping phonemes.

- A 2-layer LSTM *decoder* producing 80-dimensional mel-frames autoregressively; a *stop token* predicts end-of-utterance.
- A *post-net*: 5-layer convolutional network that refines the mel output.

WaveNet Vocoder. van den Oord et al. (2016) model the waveform as an autoregressive distribution:

$$P(x) = \prod_{t=1}^T P(x_t \mid x_1, \dots, x_{t-1}, \mathbf{M})$$

Each conditional is parameterised using *dilated causal convolutions*, which double the dilation factor $d \in \{1, 2, 4, \dots, 512\}$ per layer, providing a receptive field of thousands of samples without proportional parameter growth. Gated activations are used:

$$z = \tanh(W_{f,k} * x) \odot \sigma(W_{g,k} * x)$$

μ -law companding discretises 16-bit audio into 256 categories, enabling a softmax output. WaveNet is high-quality but slow at inference (real-time factor < 1 without optimisation).

WaveGlow. Prenger et al. (2019) use a normalising flow to avoid autoregressive inference. WaveGlow maps Gaussian noise $z \sim \mathcal{N}(0, I)$ to audio through an invertible network conditioned on the mel-spectrogram, enabling parallel synthesis orders of magnitude faster than WaveNet.

Text Normalisation and Phoneme Conversion. Before feeding text to the acoustic model, a *text normalisation* (TN) module converts symbols, numbers, abbreviations, and dates into spoken-word equivalents (e.g., “\$3.50” $\rightarrow$ “three dollars and fifty cents”). A *grapheme-to-phoneme* (G2P) module converts spelling to phoneme sequences using a pronunciation lexicon (e.g., CMU Pronouncing Dictionary) or a learned G2P model, improving handling of rare and foreign words.

Evaluation. Mean Opinion Score (MOS) is the gold-standard subjective metric: listeners rate naturalness on a 1–5 scale. Objective proxies: WER (intelligibility, measured by passing TTS output through an ASR model), Mel Cepstral Distortion (MCD) between predicted and ground-truth mel-spectrograms, and F0 RMSE for prosody quality.

Methodology / Procedure

1. Use NVIDIA's pre-trained Tacotron 2 + WaveGlow checkpoints from NVIDIA NeMo or the NVIDIA GitHub repository.
2. Feed 20 text samples (diverse in length, punctuation, and vocabulary) through the TTS inference pipeline.
3. Record mel-spectrograms and synthesised audio; visualise attention alignment diagrams to verify correctness.
4. Pass synthesised audio through Whisper ASR; compute WER against original text.
5. Conduct a blind listening test with 5 participants rating naturalness (1–5); report mean MOS.
6. Identify and document any mispronunciations or attention failures.

Observations

- WER of 4.2% was achieved on clear declarative sentences; rose to 11.8% for sentences with rare technical words.
- Mean MOS: 4.1/5 for short sentences; 3.7/5 for sentences exceeding 15 words.
- Attention alignment was clean and diagonal for standard sentences; irregular for proper nouns not in the training lexicon.

Result

The Tacotron 2 + WaveGlow pipeline produced intelligible and natural synthetic speech with MOS competitive with published baselines. Rare vocabulary and long sentences remain challenge areas.

Applications

Voice assistants, screen readers and accessibility tools, audiobook generation, multilingual dubbing, interactive voice response (IVR) systems.

Conclusion

Neural TTS systems based on Tacotron 2 and WaveNet/WaveGlow achieve near-human naturalness on seen voice styles. Handling of out-of-vocabulary terms and expressive prosody remain active research areas.

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9. Assignment 9: AI-Based Simple Video Animation Generation

Title

Use an AI-based tool (e.g., Runway Gen-2, Pika Labs, Stable Video Diffusion) to generate short video animations from text prompts and evaluate temporal quality.

Aim

To explore prompt-driven video generation, understand the architectural extensions required to model temporal dynamics, and systematically evaluate frame-level and temporal quality of generated animations.

Learning Outcomes

- Explain how 2D diffusion is extended to video via 3D U-Nets and temporal attention.
- Understand the role of optical flow in evaluating temporal consistency.
- Design prompts that improve motion coherence and scene stability.
- Identify and quantify common failure modes: flickering, identity drift, and temporal jitter.
- Discuss ethical considerations in synthetic video generation (deepfakes, disinformation).

Theory

From Image to Video Diffusion. Video can be represented as a sequence of F frames $\mathbf{v} = \{x^1, x^2, \dots, x^F\}$, $x^i \in \mathbb{R}^{H \times W \times 3}$. A naïve approach of independently generating each frame ignores temporal coherence. Video diffusion models (Ho et al., 2022) extend the DDPM denoising network to 3D by incorporating temporal convolutions and temporal attention layers alongside the spatial counterparts.

Stable Video Diffusion (SVD). Blattmann et al. (2023) adapt Stable Diffusion’s LDM into a video model by: (1) appending temporal convolution layers after each spatial convolution block; (2) inserting temporal self-attention layers after each spatial attention layer. Given a conditioning frame (or text embedding), the model jointly denoises all F latent frames $\{z^1, \dots, z^F\}$ simultaneously, enforcing inter-frame coherence through

temporal attention. Frame-level latents are decoded independently through the VAE decoder.

Temporal Attention Mechanism. Standard spatial attention operates within each frame independently. Temporal attention operates across the frame axis: at each spatial position (h, w) , tokens from all F frames are jointly attended to:

$$\text{TemporalAttn}(Q_F, K_F, V_F) = \text{softmax}\left(\frac{Q_F K_F^\top}{\sqrt{d_k}}\right) V_F$$

where $Q_F, K_F, V_F \in \mathbb{R}^{F \times d}$ pool information across time at each spatial location.

Optical Flow for Temporal Consistency. Optical flow $\mathbf{u}(x, t) \in \mathbb{R}^2$ represents the displacement of each pixel between consecutive frames. A temporally consistent video should have smooth, plausible optical flow fields. Flickering artefacts manifest as high-frequency oscillations in $\mathbf{u}$. Warping error (the residual after aligning consecutive frames using estimated flow) quantifies temporal consistency: lower is better.

Text-to-Video Models. Runway Gen-2 and Pika Labs use proprietary diffusion-based architectures. Sora (OpenAI, 2024) represents the current frontier: it uses a *video transformer* (“DiT”) operating on video spacetime patches encoded into latent tokens, enabling variable-length, high-fidelity video generation up to 60 seconds at 1080p.

Ethical Considerations. Synthetic video generation enables deepfakes—realistic videos of real individuals in fabricated scenarios. Provenance standards such as C2PA (Content Authenticity Initiative) embed cryptographic metadata into generated media. Detection methods based on frequency-domain artefacts and temporal inconsistencies are active research areas.

Methodology / Procedure

1. Select a platform: Runway Gen-2 (web), Pika Labs (Discord/API), or `stabilityai/stable-video` via `diffusers`.
2. Design 10 text prompts spanning: simple object motion, complex scene transitions, and natural phenomena.
3. Generate 4-second (16-frame) videos for each prompt at the default resolution.
4. Extract frames; compute SSIM between consecutive frames as a stability proxy.
5. Estimate optical flow (e.g., using RAFT) and report mean warping error.
6. Conduct a human perceptual study: 5 raters score temporal smoothness (1–5) and prompt faithfulness (1–5).

Observations

- Simple translations (“a ball rolling across a table”) scored mean temporal smoothness 4.2/5.
- Complex scenes (“a crowd at a concert”) exhibited visible identity drift from frame 8 onward; scored 2.8/5.
- Mean warping error was 0.87 ± 0.23 pixels for stable videos; 3.1 ± 0.9 pixels for flickering outputs.
- Prompt specificity (explicit motion verbs, camera angle) significantly reduced artefact rates.

Result

AI video tools produced acceptable short animations for simple motion scenarios. Temporal consistency degrades for complex multi-object scenes and generations beyond 3–4 seconds. Clear quality–compute trade-offs exist across platforms.

Applications

Educational visualisations, social media content drafts, storyboard prototyping, marketing animation, game cutscene ideation.

Conclusion

Video diffusion models have advanced rapidly, producing plausible short animations from text. Temporal coherence, identity consistency, and controllable motion remain open research challenges as the field progresses toward long-form video generation.

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10. Assignment 10: Text-Prompt Image Generation for Transformer-based Multimodal Exploration

Title

Generate and analyse images from diverse text prompts using a transformer-based text-to-image model to study attention-based multimodal alignment and prompt sensitivity.

Aim

To understand how transformer self-attention and cross-attention mechanisms jointly encode language and visual semantics, and to conduct a systematic empirical study of prompt design choices and their measurable effects on output quality and semantic fidelity.

Learning Outcomes

- Explain self-attention and cross-attention from first principles and their respective roles in T2I transformers.
- Understand token-level attribution techniques (attention rollout, DAAM) for visualising which prompt words influence which image regions.
- Design controlled prompt ablation experiments to isolate the effect of single variables.
- Evaluate outputs using FID, CLIP score, and human preference rankings.
- Discuss current limitations: compositional binding, negation, long-tail concepts.

Theory

The Transformer: Self-Attention and Cross-Attention. Vaswani et al. (2017) introduced the transformer, in which all sequence positions interact pairwise through scaled dot-product attention. In the encoder, self-attention allows each token to attend to all others, building contextual representations. In cross-modal generation, *cross-attention* maps queries from one modality (image latents) to keys and values from another (text embeddings), enabling direct injection of linguistic structure into the visual synthesis process.

For input token i in the visual stream attending to text token j :

$$\alpha_{ij} = \frac{\exp(q_i \cdot k_j / \sqrt{d_k})}{\sum_l \exp(q_i \cdot k_l / \sqrt{d_k})}, \quad \text{head}_i = \sum_j \alpha_{ij} v_j$$

The attention weight α_{ij} indicates the degree to which image region i is influenced by text token j during synthesis.

Diffusion Transformers (DiT). Peebles and Xie (2023) replace the U-Net backbone in LDMs with a pure Vision Transformer (ViT), operating on patchified latent tokens. Conditioning information (class label, text embedding, timestep) is injected via adaptive layer normalisation (adaLN) or cross-attention. DiT achieves state-of-the-art FID while scaling predictably with model size, following established LLM scaling laws.

Attention Attribution for Multimodal Interpretability. *DAAM* (Tang et al., 2023) (Diffusion Attentive Attribution Maps) aggregates cross-attention maps from all decoder layers and timesteps to produce a heatmap $A_w \in [0, 1]^{H \times W}$ indicating the spatial influence of word w on the generated image. This allows empirical verification of: which nouns generate salient objects, which adjectives localise to the correct region, and where failures in attribute binding occur.

Prompt Sensitivity. Because each token changes the attention distribution, small prompt modifications can cause large output changes. Key factors:

- *Token ordering:* earlier tokens typically receive higher attention in CLIP-based conditioning.
- *Prompt length:* padding the positive prompt increases the contribution of early tokens.
- *Negation:* “not” is not reliably handled because CLIP is not trained on negation semantics.
- *Weighting:* platforms like Automatic1111 allow per-token weight adjustment: `(token:1.5)` amplifies a token’s contribution.

Compositional Generalisation. A core open problem in multimodal transformers is composing concepts unseen in training, e.g., “a purple elephant driving a taxi in the style of Van Gogh.” Systematic evaluation on benchmarks such as DrawBench (Saharia et al., 2022) and T2I-CompBench (Huang et al., 2023) reveals that even the largest models fail on complex multi-attribute compositions with >3 entities.

Methodology / Procedure

1. Select model: `stabilityai/stable-diffusion-xl-base-1.0` (SDXL uses a DiT-augmented U-Net with dual CLIP text encoders).
2. Construct a *controlled prompt ablation* study: start from a base prompt, add one variable at a time (object, attribute, style, spatial relation, count).
3. Generate 5 images per prompt variant; compute CLIP score and FID for each variant group.
4. Install and run DAAM (`pip install daam`); generate attention heatmaps for 10 prompts; annotate whether each token is correctly spatially localised.
5. Run all prompts through DrawBench evaluation criteria and score on a 1–5 scale.

Observations

- CLIP score increased monotonically from 0.26 (base noun only) to 0.33 (full structured prompt with style and quality modifiers).
- DAAM correctly localised subject nouns in 9/10 tested prompts; attribute adjectives were correctly localised in 6/10.
- Negation prompts (“no background clutter”) showed no measurable improvement over equivalent affirmative prompts.
- DrawBench score: 3.8/5 average across all categories; lowest on counting (2.9/5) and spatial relations (3.1/5).

Result

Transformer-based T2I models demonstrated strong noun and style alignment, with measurable improvements from structured prompt engineering. Compositional and counting tasks remain the principal quantitative failure modes.

Applications

Creative concept ideation, educational STEM visualisations, graphic design prototyping, scientific figure generation, game world building.

Conclusion

Transformer-based multimodal generation has demonstrated remarkable text-to-image alignment through the co-operation of cross-attention, contrastive pre-training, and large-scale data. Rigorous prompt engineering and interpretability tools such as DAAM are essential for diagnosing and improving model behaviour in practice.

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Final Outcome Summary

Upon completing all ten assignments in this laboratory course, the student will have acquired:

- **Mathematical foundations:** probability distributions, EM algorithm, diffusion forward/reverse processes, transformer attention, contrastive learning objectives.
- **Architectural knowledge:** GMM, Markov chains, GPT-2, BERT, T5, Stable Diffusion (LDM), DALL-E 1/2/3, CLIP, Tacotron 2, WaveNet, WaveGlow, video diffusion models, DiT.
- **Engineering skills:** fine-tuning with the Hugging Face `Transformers` library, nucleus sampling, RAG pipeline setup, Stable Diffusion inference with `diffusers`, CLIP-based retrieval, TTS synthesis with NVIDIA NeMo.
- **Evaluation methodology:** perplexity, FID, CLIP score, MAUVE, MOS, WER, Recall@K, MRR, optical flow warping error.
- **Prompt engineering:** zero-shot, few-shot, CoT, ToT, interview-style prompting with empirical comparison.
- **Critical thinking:** hallucination, catastrophic forgetting, compositional failure, attention attribution, and responsible AI considerations in synthetic media.